

Review via Linear Regression

02-680: Essentials of Mathematics and Statistics

September 3, 2026

The **linear regression** problem is as follows (we covered this briefly in a previous topic (Independence), but with a slightly different formula):

x — Independent variable[s] (what we control)

y — Dependent variable[s] (the measured values)

β_s — slope

β_0 — intercept

Some examples of regression in biology:

Independent	Dependent
Expression of genes expressed in immune response	asthma severity
Expression of regulator genes	Expression of target genes
Clinical variables	Insulin level
...	...

Lets assume we have some $x, y \in \mathbb{R}^n$, for n samples where each (x_i, y_i) are corresponding. Then

$$y = x^T \beta_s + \beta_0$$

To simplify things lets define: $X = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}$ and $\beta = \begin{bmatrix} \beta_s \\ \beta_0 \end{bmatrix}$, then the equation above

becomes simply:

$$y = X\beta.$$

Recall (possibly) from earlier that we want to find β that minimizes the $\|y - X\beta\|_2^2$, or the square of the L_2 -norm.

$$\begin{aligned}
\|y - X\beta\|_2^2 &= (y - X\beta)^T (y - X\beta) \\
&= (y^T - \beta^T X^T)(y - X\beta) \\
&= y^T y + \beta^T X^T X \beta - 2\beta^T X^T y
\end{aligned}$$

To find the critical point of the curve (i.e. the minimum) we need

$$\begin{aligned}
\frac{d}{d\beta} y^T y + \beta^T X^T X \beta - 2\beta^T X^T y &= 0 \\
\frac{d}{d\beta} \beta^T X^T X \beta - \frac{d}{d\beta} 2\beta^T X^T y &= 0 \\
2X^T X \beta - 2X^T y &= 0 \\
2X^T X \beta &= 2X^T y \\
\beta &= (X^T X)^{-1} X^T y
\end{aligned}$$

What if I want to know the probability of some y_i given x_i ? (so $p(y_i | x_i)$)

Assuming its some natural process, we can assume

$$p(y_i | x_i) \sim \mathcal{N}(\beta_0 + \beta_s x_i, \sigma^2).$$

or equivalently

$$y = \beta_0 + \beta_s x_i + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2).$$

So given the data, we can actually use MLE to find β !

$$\begin{aligned}
\ln p(\mathcal{D} | \beta, \sigma) &= \ln \left[\prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2} (y_i - x_i \beta)^2} \right] \\
&= \sum_{i=1}^n \ln \left[\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2} (y_i - x_i \beta)^2} \right] \\
&= \frac{-n}{2} \ln \sigma^2 + \sum_{i=1}^n \ln e^{-\frac{1}{2\sigma^2} (y_i - x_i \beta)^2} \\
&= \frac{-n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i \beta)^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial \beta} \left(-\frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i \beta)^2 \right) &= \frac{\partial}{\partial \beta} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i \beta)^2 \\
&= \frac{\partial}{\partial \beta} - \frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta) \\
&= -\frac{1}{2\sigma^2} \frac{\partial}{\partial \beta} (Y^T Y - 2Y^T X\beta + \beta^T X^T X\beta) \\
&= \dots
\end{aligned}$$

$\hat{\beta}_{MLE} = (X^T X)^{-1} X^T y$, that's just Least Squares!

But we don't know σ , that's actually, what we want to know because it models how uncertain we are with any given output.

$$\frac{\partial}{\partial \sigma^2} \left(-\frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i \beta)^2 \right) = \frac{-n}{2\sigma^2} + \frac{1}{2} (\sigma^2)^{-2} \sum_{i=1}^n (y_i - x_i \beta)^2$$

$$\hat{\sigma}_{MLE}^2 = \frac{\sum_{i=1}^n (y_i - x_i \hat{\beta})^2}{n}$$

With these we can determine for some new $\tilde{y}$ and $\tilde{x}$ what $p(\tilde{y} | \tilde{x})$ is.